

3 BEAR Probe

We base our evaluation on using an LM’s inherent ability to estimate the log-likelihood of a given sentence. Our main idea is to restrict the space of possible answers for each relation instance and create a set of options that are then ranked by their log-likelihood values.

3.1 Ranking Options using Log-Likelihood

Our approach requires a dataset of $\langle s, r, o \rangle$ relation instances, where for each relation r there exists (at least) one template t and a set of answer options a_i with $i \in \{1, \dots, k\}$ that includes the correct answer.

Creating options to rank. For each relation instance, we create k natural language statements using the template, instance’s subject s , and each of the possible relation’s objects a_i as parts of a textual statement.

Figure 3 illustrates this process for the example relation instance $\langle \text{“Uganda”}, \text{HAS-CAPITAL}, \text{“Kampala”} \rangle$ and the template “The capital of [X] is [Y]”. This example’s set of potential answers is [“Kampala”, “Thimphu”, “Buenos Aires”, “Bandar Seri Begawan”]. For each potential answer, we create a separate statement.

Predicting log-likelihood. For each generated statement, we predict the log-likelihood score $\log \hat{p}(a|t)$. As the template is the same for each of the answer options (i.e., $\hat{p}(t)$ is constant) and we are only interested in ranking them, it is sufficient to compute the log-likelihood for the entire sentence:

$$\log \hat{p}(a_i|t) = \log \hat{p}(a_i, t) - \log \hat{p}(t)$$

Since causal LMs are trained to predict a log-likelihood of each token given the previous context, the log-likelihood of a sentence is simply the sum of the log-likelihoods of each token.

Log-likelihood in masked LMs. A sentence-level log-likelihood is not clearly defined for an LM trained using the masked language modeling objective. However, Salazar et al. (2020) and Kauf and Ivanova (2023) offer two variants of how to retrieve a pseudo log-likelihood score for a given text. Both approaches use multiple forward passes. Salazar et al. (2020) simply mask each token once while keeping the remaining context unmasked. Kauf and Ivanova (2023) improve on this by additionally masking all tokens right to the current

token belonging to the same word. This approach fixes the issue of assigning disproportionate likelihoods to multi-token words. We use the latter in our approach.

Ranking the results. Finally, the statements are ranked by their log-likelihood scores. This is illustrated in Figure 3 (right-hand side).

3.2 Evaluation Metric

To evaluate the amount of knowledge encoded in each model, we score whether the top-ranked statement is the correct answer for each relation instance. Previous work (such as Petroni et al., 2019) additionally considered answers with higher ranks. The mean precision $P@k$ (for a given rank k) is commonly reported.

However, given that we evaluate on a constrained answers space, we believe the first rank to be sufficient. In the case of $k = 1$, this is identical to the accuracy metric. We report the average over all relation instances in our evaluation data as the BEAR score.

4 BEAR Dataset

Our proposed probing approach requires a dataset with a restricted answer space. Following the analysis in Section 2 we additionally desire (1) the answer space to be balanced, (2) a single correct answer per relation instance, (3) a balanced number of instances per relation, and (4) a focus on knowledge that could reasonably be found in corpora other than Wikipedia.

4.1 Selecting Relations

We use the 234 relations of KAMEL as a starting point and manually remove two-thirds of these. This curation process was conducted independently by two researchers (authors of this paper), and disagreeing judgments were discussed in detail to reach a final decision. The most common reasons for excluding a relation were:

- A relation (after filtering) had too few objects with a desired number of instances (i.e., given the relation’s statistics, it was impossible to build a balanced answer space within our constraints).
- We expect the relational information to undergo significant changes (hence, it is not time-invariant). For example, we assume the RESIDENCE relation, linking an individual to their

place of residence, to be highly susceptible to change over time. Including such relations without accounting for their temporal context would unfairly benefit LMs trained on datasets from the same time period as the evaluation data.

- A relation with many instances that have incomplete object listings in Wikidata since this may cause correct answers to be counted as errors. For example, the MADE-FROM-MATERIAL relation, which connects an object and the material it is made of, often contains only a few primary components as objects.
- Overly diverse subject or object space, making the semantics of the relation overly broad and impairing our ability to design meaningful templates. For instance, the relation COUNTRY connects various entity types, such as events, ships, roles, websites, TLDs, codes of standards, and many more categories, to a country.

As a result of this process, we selected 78 relations for inclusion in the larger variant of BEAR.

4.2 Selecting Relation Instances

For the selected relations, we retrieve relation instances from Wikidata⁴ and employ a number of filtering steps to ensure that only instances meeting our desiderata remain.

Filtering subjects and objects. We first filter down the space of eligible subjects and objects. We remove all Wikidata entities (i.e., subjects and objects) that do not have an English label. Following prior work (Poerner et al., 2020), we additionally remove all subjects with overly revealing entity names. For example, predicting the name of the company that produced the “Apple Watch” is straightforward since the correct answer (i.e., “Apple”) is part of the subject (i.e., “Apple Watch”). The similarity is computed via the token overlap and fuzzy string matching (Bachmann, 2023).

Ensuring a coherent answer space. Even in our curated set of relations, some relation instances are connected to outlier object types. For instance, the HEAD-OF-GOVERNMENT relation, which typically connects a country to a specific named person (e.g.,

“Joe Biden”), would in some cases connect to a job title instead (e.g., “president”).

To increase coherence and ensure that our templates are meaningful, we utilized GPT4 (OpenAI, 2023) to flag answers which stand out (see Figure 11 in Appendix F for the template that was used) and decided on a case-by-case basis whether to accept these changes. This process also helped us check the relations for potential issues.

Sampling a balanced dataset. For this initial set of entities, we sampled the remaining relation instances such that (1) each relation has a uniform distribution of objects in the answer space, (2) each relation has approximately the same number of instances overall, and (3) there is no overlap between various entity names within a relation. During sampling, we gave preference to Wikidata entities with Wikipedia pages in multiple languages to focus on well-known entities that might reasonably be found in corpora outside of Wikipedia.

This process yields a total of 40,916 instances for our 78 relations.

4.3 Further Refinement

The large variant of the dataset consists of a broad range of subjects and captures a considerable amount of relational knowledge found in Wikidata. However, considering the extensive scope of the probe and the evaluation scheme employed, the time required for a complete assessment is considerable. Hence, we further developed a subset of the full probe designed to be more time-efficient.

We imposed a constraint on the answer space, setting an upper limit of 25 answers for 1:N relations. This upper limit was set to keep the task challenging while making the evaluation more efficient. To further refine this subset, we applied a filtering criterion based on entity popularity, excluding entities (objects and subjects) linking to Wikipedia pages with fewer than 10,000 page views (between 2016 and 2023). This step ensures the relevance and recognition of the included entities. Subsequently, through an iterative process, we identified optimal configurations that achieve a balanced answer space with a target of approximately 150 instances. Configurations deemed infeasible were systematically removed (in total, 18 relations, leaving 46 relations). The remaining possible answers were then sorted based on their subjects’ median page view count and checked in a manual review. During this review, we examined the answers and

⁴We use the JSON dump of Wikidata of January 3rd, 2022 (Wikidata contributors, 2022) which is available as a torrent under a CC BY-SA license.

their respective subject samples, leading to the exclusion of problematic subject instances. We then included subjects with the highest page view counts.

In the case of 1-1 relations, the process was similar but using a limit of 60 answers/instances per relation. Additionally, we selected a random sample of instances rather than ranking them by popularity for these relations.

In the following sections of the paper, we will use “BEAR” to refer to this subset and “BEAR_{big}” to denote the full probe.

4.4 Templates

We create three templates for each relation to better safeguard against template-specific biases. We source the initial templates from the existing LAMA dataset, utilize GPT4 to create additional ones (the used prompt can be found in Figure 12 in Appendix F), and manually select those that best match our subjects and answer spaces. Finally, we query GPT4 with each of the templates applied to 5 random subject-object pairs from each relation to check for linguistic correctness (the used prompt can be found in Figure 13 in Appendix F).

4.5 Resulting Dataset

The final dataset consists of 60 relations and 7,731 items. Most of these relations are 1:N, each featuring a restricted answer space ranging from 5 to 25 possible answers, with an average of roughly 23 answers. The answer space is also balanced so that each answer appears the same number of times across all instances. Each answer has between 6 and 30 instances, with an average of approximately 6.5 instances per answer. The dataset also contains 14 1:1 relations that contain only one instance per answer.

For a detailed comparison of these statistics with LAMA and KAMEL, see Table 2.

5 Experiments

We present an experimental evaluation using the BEAR probe on a selection of LMs, compare our results to earlier probes, and discuss the results.

Compared LMs. We compare a total of 22 LMs, as listed in Table 3: This includes 6 masked LMs from the BERT, RoBERTa, XLM-RoBERTa families, each in their base and large variants. Additionally, we include 16 causal LMs from the GPT

Dataset	LAMA	KAMEL	BEAR _{big}	BEAR
# Instances	31,479	46,800	40,916	7,731
# Relations	41	234	78	60
Literals	no	yes	no	no
1:1 Relations	0		14	14
N:1 Relations	7		64	46
N:M Relations	34		0	0
N:M Instances	1,035	4,296	0	0
Avg. Instances per Relation	830.2	1,400 ⁵	596.9	149.8

Table 2: Descriptive dataset statistics: BEAR compared to LAMA (T-REx subset) and KAMEL (figures for KAMEL and LAMA from Kalo and Fichtel, 2022). Avg. Instances per Relation only includes relations with more than one instance per answer.

and OPT families, along with newer models such as Llama2, Gemini, and Mistral. We assess 5 different sizes for the OPT models to examine the relationship between the BEAR score and increasing model sizes.

BEAR score. We compute the BEAR score for each of the three template options per relation individually and report the average across templates as well as the standard deviation.

5.1 Main Results

Table 3 lists the results for all LMs in consideration. We present the overall BEAR score and the scores for the subsets of 1:1 and N:1 relations only. We find that scores are generally low for all models, highlighting the challenging nature of our benchmark, as it queries factual information with strong detractors. In addition, we make a number of observations:

BEAR scores are higher for larger LMs. In line with our expectations, we find that larger models consistently outperform their smaller counterparts. For a better illustration, we present a plot of accuracy against model size in Figure 4. This trend of steady accuracy improvement with increasing model size is evident across all tested model families. Interestingly, the smallest change is observed among BERT models, where the performance of bert-base-cased and bert-large-cased across all of the relations is roughly on par. We further observe that recent models generally achieve higher performance, with Mistral-7B-v0.1 (2023) and gemma-7b (2024) surpassing the Llama2-7b (2023) model (when compared at identical parameter counts), which itself significantly outperforms the opt-6.7b model from 2022.

⁵1,000 train samples, and 200 each for dev & test

Model	Type	# params	BEAR	BEAR _{1:1}	BEAR _{1:N}
Llama-2-13b-hf	CLM	13b	66.9% $\pm$ 1.0%	66.5% $\pm$ 1.6%	67.0% $\pm$ 1.1%
Mistral-7B-v0.1	CLM	7.0b	65.4% $\pm$ 1.1%	64.5% $\pm$ 1.2%	65.5% $\pm$ 1.1%
gemma-7b	CLM	7.0b	63.7% $\pm$ 1.3%	63.5% $\pm$ 0.7%	63.8% $\pm$ 1.4%
Llama-2-7b-hf	CLM	7.0b	62.4% $\pm$ 1.3%	62.2% $\pm$ 1.1%	62.4% $\pm$ 1.3%
gemma-2b	CLM	2.0b	51.5% $\pm$ 1.0%	53.1% $\pm$ 1.3%	51.3% $\pm$ 1.0%
opt-30b	CLM	30b	47.9% $\pm$ 0.5%	45.8% $\pm$ 1.0%	48.2% $\pm$ 0.6%
opt-13b	CLM	13b	45.4% $\pm$ 0.8%	43.5% $\pm$ 2.1%	45.7% $\pm$ 0.6%
opt-6.7b	CLM	6.7b	43.8% $\pm$ 1.1%	42.5% $\pm$ 1.0%	43.9% $\pm$ 1.2%
opt-2.7b	CLM	2.7b	37.3% $\pm$ 0.9%	35.6% $\pm$ 0.7%	37.5% $\pm$ 1.0%
opt-1.3b	CLM	1.3b	31.5% $\pm$ 0.8%	31.3% $\pm$ 0.6%	31.5% $\pm$ 0.9%
gpt2-xl	CLM	1.6b	26.2% $\pm$ 0.7%	24.1% $\pm$ 1.6%	26.5% $\pm$ 0.6%
gpt2-large	CLM	812M	22.2% $\pm$ 0.6%	20.1% $\pm$ 1.8%	22.5% $\pm$ 0.5%
roberta-large	MLM	355M	21.5% $\pm$ 0.8%	22.0% $\pm$ 1.1%	21.5% $\pm$ 0.8%
bert-large-cased	MLM	335M	19.9% $\pm$ 0.5%	16.6% $\pm$ 1.0%	20.3% $\pm$ 0.5%
opt-350m	CLM	350M	19.6% $\pm$ 0.6%	18.6% $\pm$ 1.2%	19.7% $\pm$ 0.6%
gpt2-medium	CLM	355M	19.0% $\pm$ 0.8%	16.0% $\pm$ 2.6%	19.4% $\pm$ 0.6%
bert-base-cased	MLM	109M	18.4% $\pm$ 0.4%	15.0% $\pm$ 1.1%	18.8% $\pm$ 0.4%
roberta-base	MLM	125M	16.4% $\pm$ 0.7%	15.8% $\pm$ 1.8%	16.5% $\pm$ 0.8%
opt-125m	CLM	125M	16.4% $\pm$ 0.5%	14.0% $\pm$ 1.3%	16.7% $\pm$ 0.4%
xlm-roberta-large	MLM	561M	14.3% $\pm$ 0.3%	14.9% $\pm$ 1.7%	14.3% $\pm$ 0.5%
gpt2	CLM	137M	13.5% $\pm$ 0.8%	9.4% $\pm$ 2.1%	14.0% $\pm$ 0.7%
xlm-roberta-base	MLM	279M	11.4% $\pm$ 0.2%	11.4% $\pm$ 1.1%	11.4% $\pm$ 0.2%
Random Baseline	-	-	4.7%	1.7%	5.1%

Table 3: Models investigated in this work (Devlin et al., 2019; Jiang et al., 2023; Gemini-Team, 2023; Liu et al., 2019; Radford et al., 2019; Touvron et al., 2023; Zhang et al., 2022) sorted by their BEAR score (weighted average over all relations) and as the mean over all templates (with the standard error).

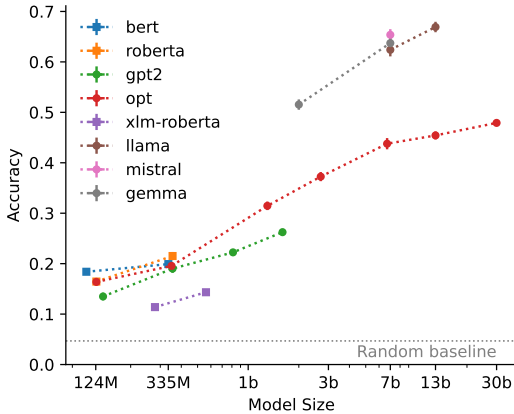


Figure 4: Probing scores of different models on BEAR. Model size is represented on a log scale.

Better BEAR scores for masked LMs. When comparing models by their parameter count, we note a slight advantage of masked over causal LMs. This may indicate that the masked language modeling objective, encouraging deep bidirectionality, is more effective in capturing factual knowledge.

Impact of multilingual training data. We note that the two XLM-RoBERTa models are among the lowest-scoring models in the benchmark. We hypothesize that this diminished performance of the XLM models may stem from its pre-training on multilingual corpora and a focus of BEAR on English-language entities.

Impact of templates. We further evaluate the im-

part of template choice on the BEAR score. A full analysis of all relations is provided in Figure A in the Appendix.

In line with the observation of Elazar et al. (2021), we find that LMs are sensitive to how they are queried. For instance, in the CAPITAL-OF relation, the accuracy of bert-base-cased drops approximately by 80% when using “[Y] has its governmental seat in [X]” instead of “The capital of [X] is [Y].”. While some states have their government seat in a different location than their official capital city, this fact alone cannot account for the significant disparity in observed performance. Interestingly, we don’t observe any drop in performance between these templates in the case of other models (e.g., Llama2 models). We hypothesize that this might be due to BERT’s primary training on Wikipedia, which has limited exposure to diverse writing styles.

5.2 Comparison to LAMA

To compare BEAR and LAMA, we consider their common subset of relations and utilize our proposed log-likelihood based evaluation technique. See Figure 5 for an illustration. We find BEAR to be a more challenging probe compared to T-REx, the dataset used in LAMA. We attribute this to several design choices, namely the balanced answer space and the absence of overly informative entity names, forcing LMs to rely solely on the knowl-